AMACE

Autonomous Multi-Stratum Anomaly Correlation Engine

A Zero-Configuration Architecture for Counterfactually-Validated Causal Discovery.

Combinatorial Explosion Bound: $O(N^2 \times K^D) > 10^{18}$

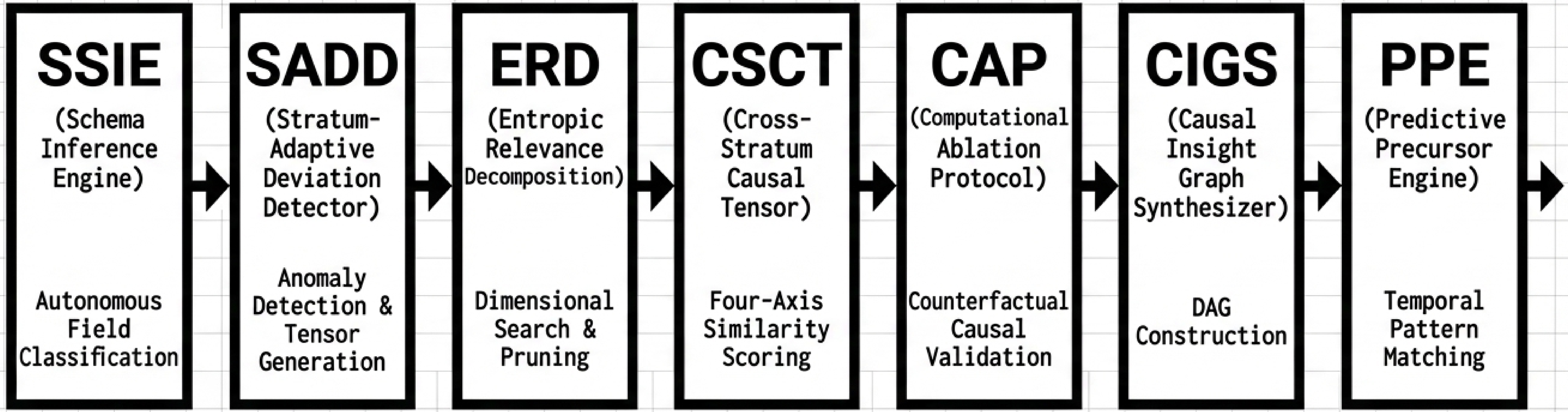
For typical production systems (N=1000 anomalies, D=8 dimensions, K=50 cardinality), exhaustive search across strata is computationally intractable.

The Cross-Stratum Blind Spot

Observation Stratum – S_0	Unprocessed event data 10^6–10^{12} events/day	No semantic interpretation.
Representation Stratum – S_1	Derived statistical metrics Moderate volume	Engineered knowledge.
Inference Stratum – S_2	ML/Statistical model outputs Low volume	Predictive intelligence.

System deviations cascade across S_0, S_1, and S_2 simultaneously. Existing architecture cannot bridge the temporal and structural schema gaps without exponential resource collapse.

The AMACE Pipeline Architecture



Module 1: Stratum Schema Inference Engine (SSIE)

Zero-Configuration Entropy-Based Field Classification

$$P_{entity}(f) = \text{sigmoid}\left(\alpha_1 \cdot H(f) + \alpha_2 \cdot \frac{|\text{unique}(f)|}{|f|} + \alpha_3 \cdot \text{uniformity}(f)\right)$$

$$P_{timestamp}(f) = \text{sigmoid}(\beta_1 \cdot \text{monotonicity}(f) + \beta_2 \cdot \text{periodicity}(f) + \beta_3 \cdot \text{datetime\_parseable}(f))$$

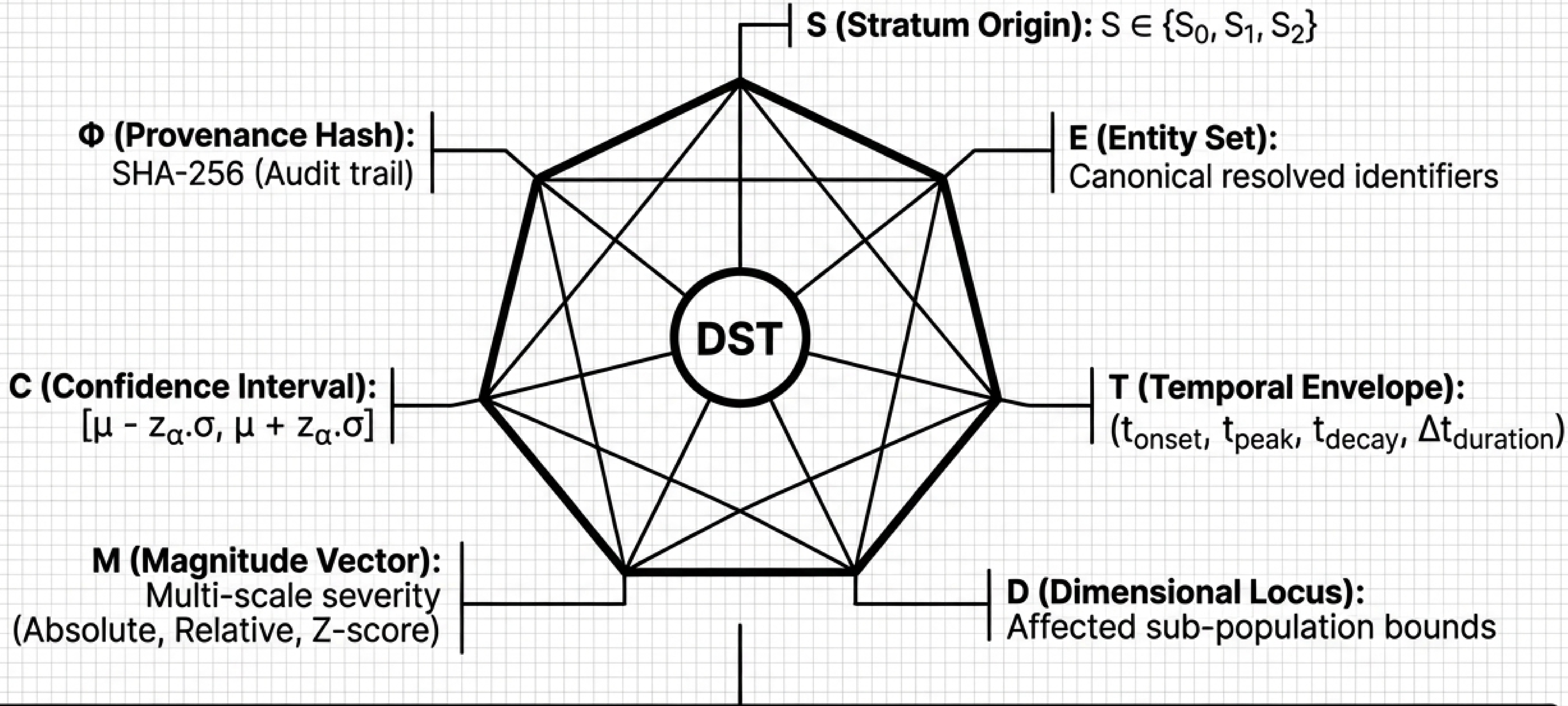
$$P_{dimension}(f) = \text{sigmoid}(\gamma_1 \cdot (1 - |\text{unique}(f)|/|f|) + \gamma_2 \cdot \text{categorical}(f) + \gamma_3 \cdot \text{temporal\_stability}(f))$$

$$P_{metric}(f) = \text{sigmoid}(\delta_1 \cdot \text{numeric}(f) + \delta_2 \cdot \text{continuity}(f) + \delta_3 \cdot \text{temporal\_variance}(f))$$

Streaming Computation Bound: $O(F \times S)$
Computes probabilities autonomously using Shannon entropy ($H(f)$) and cardinality ratios.
No iterative optimization or human definitions required.

Module 2: Stratum-Adaptive Deviation Detector (SADD)

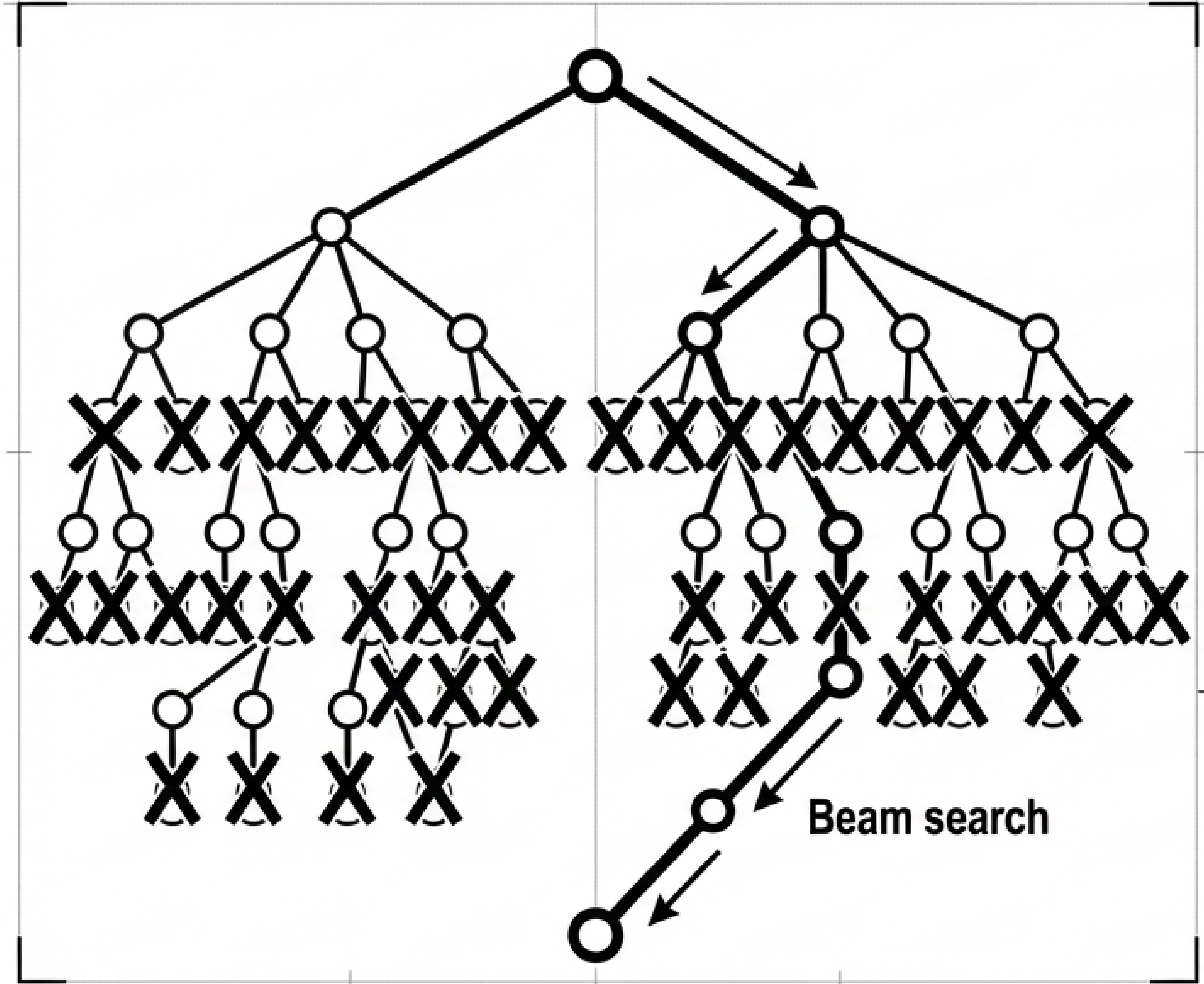
The Universal Deviation Signature Tensor (DST)

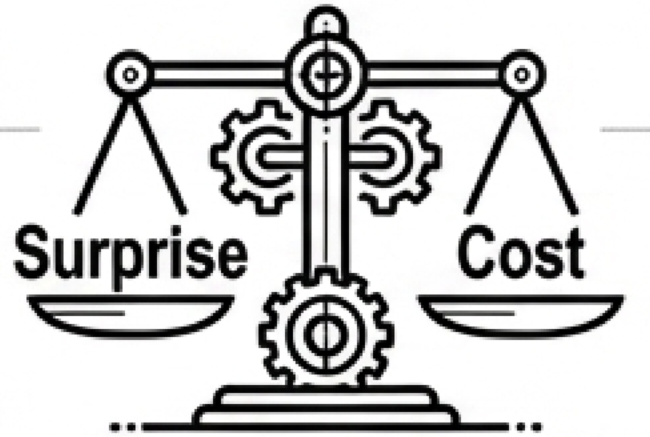


Detection Mechanics: Stratum-specific triggers standardize into the DST via Adaptive Z-scores (Median Absolute Deviation) for S_0, Population Stability Indices (PSI) for S_1, and Prediction Interval Monitoring for S_2.

Module 3: Entropic Relevance Decomposition (ERD)

Dimensional Search via Entropic Pruning





Algorithm Objective: Identify the most informationally relevant dimensional slice (σ*) maximizing:

$$\mathrm{ERD}(\sigma) = \mathrm{Surprise}(\sigma) - \mathrm{Cost}(\sigma)$$

The Pruning Logic:

$$\mathrm{Surprise}(\sigma) = \log_2\left(\frac{m_{slice}}{m_{parent}}\right) \times |E_{slice}|$$ (Information gain)

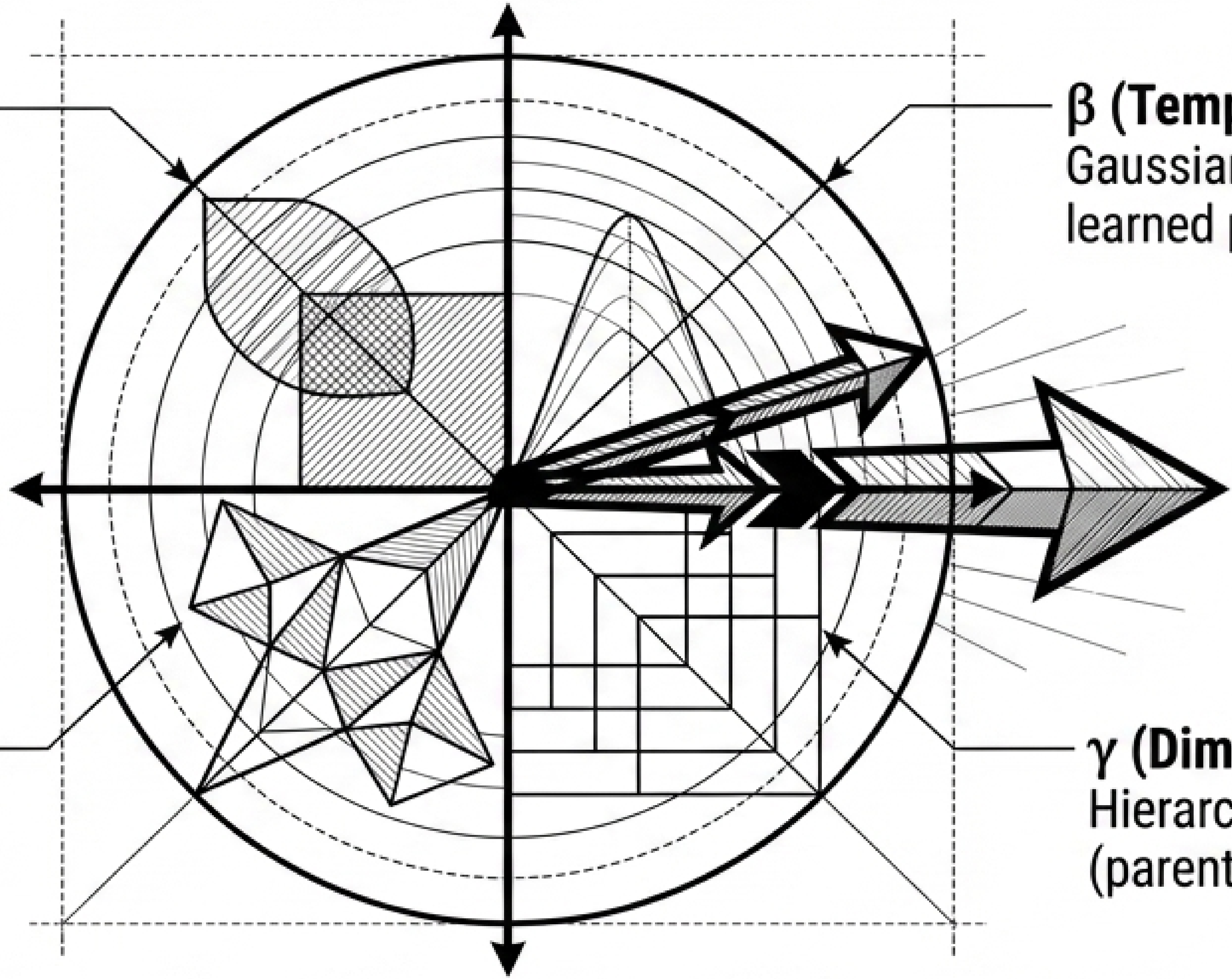
$$\mathrm{Cost}(\sigma) = \mathrm{depth}(\sigma) \times \log_2(\mathrm{cardinality}(\sigma))$$ (Description length)

IF Cost(σ) > Surprise(σ) **THEN** Drop Branch

Complexity Breakthrough: Beam search limits exploration width, achieving computational bounds of $O(w \times D \times K_{\max})$. Reduces combinations from $O(10^{16})$ to ~40,000 evaluations.

Module 4: Cross-Stratum Causal Tensor (CSCT)

Moving Beyond Ad-Hoc Correlation to Four-Axis Similarity



α (Entity Coincidence): Surprise-adjusted Jaccard similarity.

β (Temporal Precedence): Gaussian kernel centered on learned propagation delays (μ_{delay}).

δ (Semantic Affinity): Self-trained metric embeddings via contrastive learning.

γ (Dimensional Coherence): Hierarchical dimension matching (parent-child relationship).

Spectral Normalization Equation: $\mathrm{CSCT}(A,B) = \left\| \sum_i w_i \times \sigma_i(A,B) \times \hat{e}_i \right\|$

Weights (w_i) adapt dynamically based on signal characteristics (e.g., sudden onset vs. gradual drift).

Module 5: Computational Ablation Protocol (CAP)

Counterfactual Causal Validation

Gate 1: Exclusivity Ablation

Test: Remove DST_A. Does DST_B disappear?

Criterion:

$$\frac{|m_{\text{B\_original}} - m_{\text{B\_ablated}}|}{m_{\text{B\_original}}} > \theta_{\text{exclusivity}}$$

Gate 2: Sufficiency Ablation

Test: Use only DST_A to predict DST_B.

Criterion:

$$\frac{|m_{\text{B\_predicted}} - m_{\text{B\_observed}}|}{m_{\text{B\_observed}}} < \theta_{\text{sufficiency}}$$

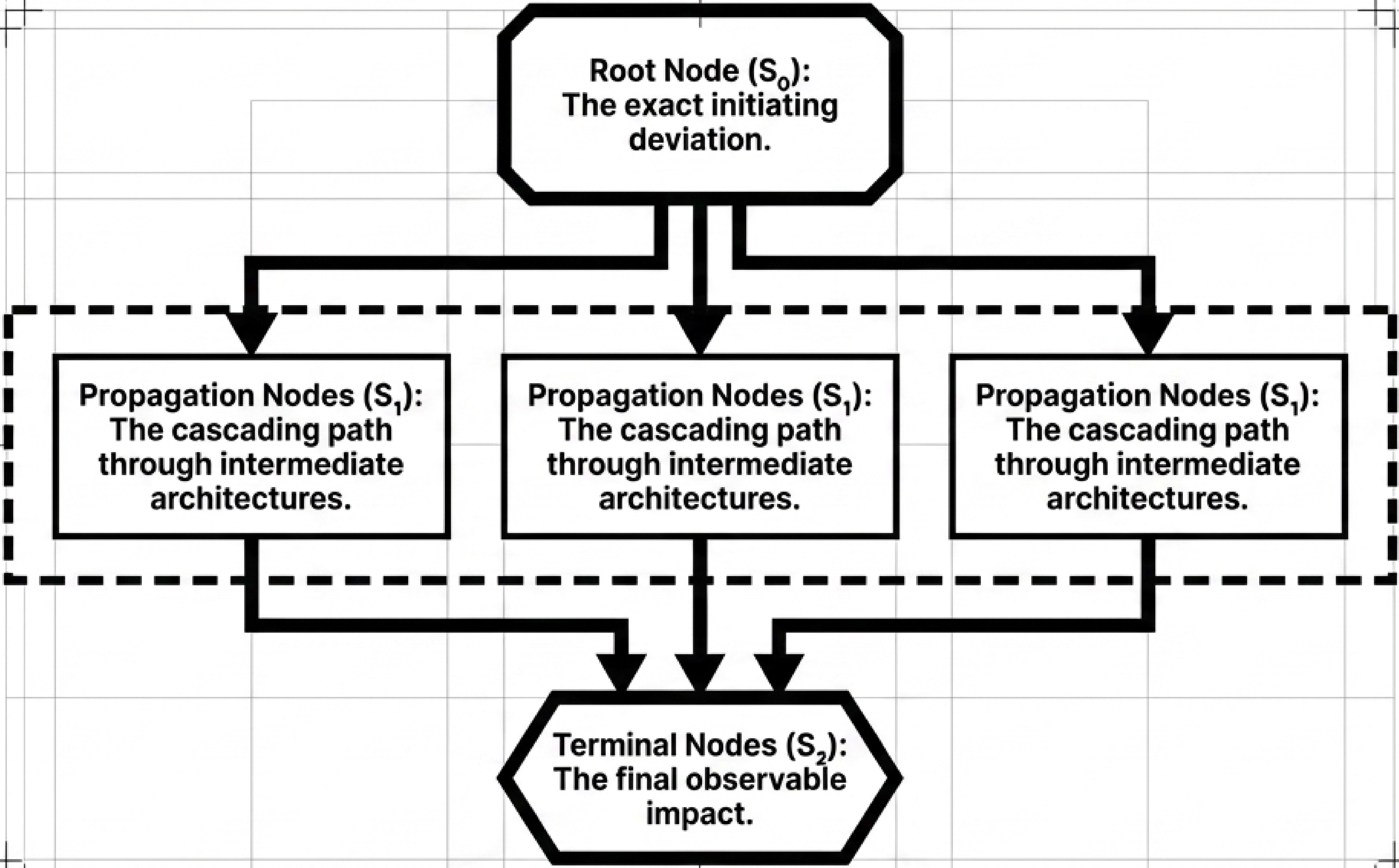
Gate 3: Minimality Ablation

Test: Iteratively remove DSTs in N>2 clusters to find the minimal sufficient set, preventing over-attribution.

Statistical Rigor Label: Validated via 1,000 bootstrap iterations holding Type I error bounds strictly at $\alpha = 0.05$.

Module 6: Causal Insight Graph Synthesizer (CIGS)

The Final Counterfactually-Validated Truth

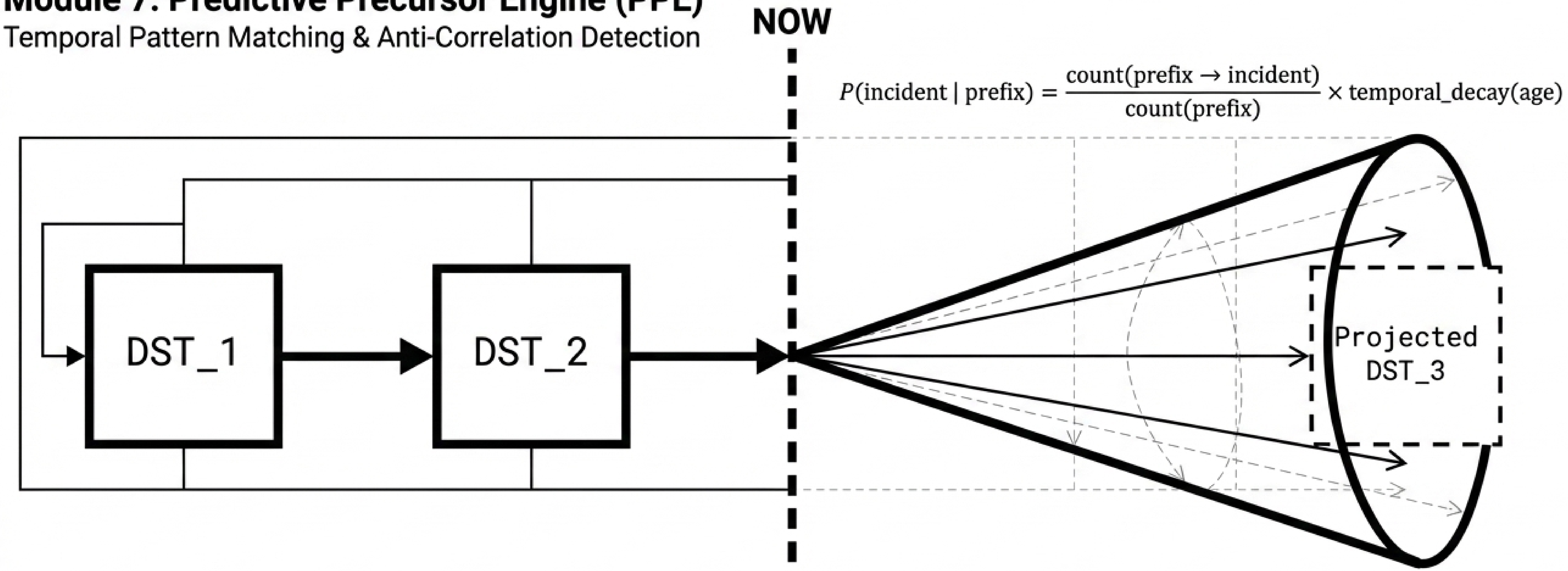


Structured Output Data

```
[
  - Assembles validated
    tensors into an
    unambiguous causal
    cascade.

  - Outputs exact affected
    entity intersections.

  - Aggregate confidence
    scores.

  - Cryptographic
    provenance hash chains
    for absolute auditabi-
    lity.
]
```

Module 7: Predictive Precursor Engine (PPE)

Temporal Pattern Matching & Anti-Correlation Detection



$$P(\text{incident} \mid \text{prefix}) = \frac{\text{count}(\text{prefix} \to \text{incident})}{\text{count}(\text{prefix})} \times \text{temporal\_decay}(\text{age})$$

The Precursor Sequence Library (PSL)	Anti-Correlation Note
When a new DST matches the first *k* elements of a known sequence length *n*, the system issues automated probability warnings for the impending terminus.	Detects missing expected correlations (e.g., latency spikes without autoscaler triggers), uncovering latent configuration errors or silent infrastructure failures.

NotebookLM

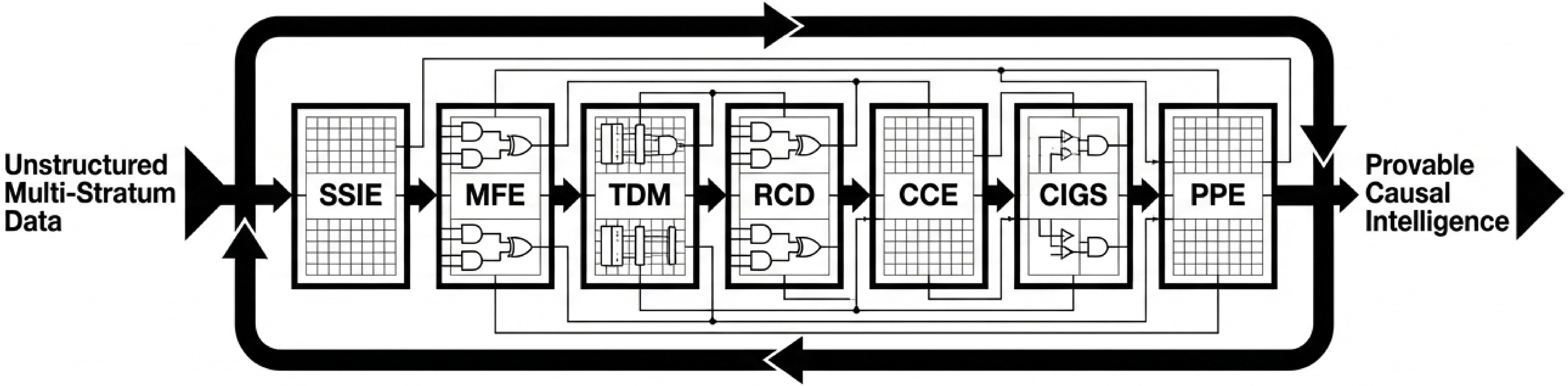
Architectural Synthesis

The Mathematical Paradigm Shift in Causal Discovery

Computational Bounds Comparison

~~`Before AMACE:` $O(N^2 \times K^D) > 10^{18}$
(Combinatorial Blind Spot)~~

$\boxed{\texttt{With AMACE: } O(N \times \log(N) \times k \times \log(D)) \text{ (Provable Convergence)}}$



"Transforming unconfigured, heterogeneous noise across disparate strata into counterfactually-validated truth. A domain-agnostic engine that requires zero human configuration, fundamentally altering the bounds of system observability."